INTEGRATED BUILDING ENVELOPE DESIGN: A BIM BASED WORKFLOW FOR ACHIEVING SUSTAINABILITY

1.Introduction

The building envelope is a critical interface between the internal environment of a building and external climatic conditions. It influences heat gain and loss, daylight availability, solar exposure, natural ventilation, thermal comfort and consequently the overall energy performance of a building. Conventional envelope design often prioritizes architectural appearance and functional requirements, while environmental performance is evaluated at later stages of the design process.

Building Information Modelling (BIM) provides an opportunity to integrate environmental performance considerations into the architectural design workflow. BIM models can provide geometric and material information that can be evaluated using performance-analysis tools. This enables designers to study how envelope parameters such as orientation, glazing, shading, thermal properties and façade configuration affect building performance during the design process.

However, the availability of multiple BIM-based simulation and analysis tools does not necessarily establish a clear workflow for architects to integrate envelope design decisions with sustainability objectives. There is therefore a need to investigate how BIM can support an integrated envelope-design process in which environmental performance is assessed alongside architectural design.

2.Background of study

Building sustainability increasingly requires environmental considerations to be incorporated during the early design stages rather than treated as an assessment conducted after the building has been designed. The building envelope plays an important role because it regulates the exchange of heat, light and air between the building and its surroundings.

BIM can provide a common digital environment in which geometric and material information can be generated, modified and transferred to analysis platforms. Architects may use different software for modelling, environmental simulation and sustainability assessment, resulting in repeated modelling, loss of information and difficulty in translating simulation results into practical design decisions.

3.Problem statement

Although BIM and environmental-analysis tools provide opportunities to evaluate building performance, building-envelope decisions are often made independently from environmental performance analysis. Therefore, the research seeks to establish a structured BIM-based workflow that can assist architects in making environmentally informed building-envelope decisions during the design process.

4.Research Aim & Objectives

"To develop a BIM-based workflow for building-envelope design that supports the evaluation and optimization of envelope strategies towards achieving building sustainability."

Objectives:

1. To study the relationship between building-envelope design parameters and environmental performance.
2. To identify key envelope strategies that contribute to building sustainability.
3. To investigate the capabilities of selected BIM-based environmental-analysis tools for evaluating building-envelope performance.
4. To analyse selected sustainable buildings to identify effective envelope strategies and their relationship with environmental performance.
5. To develop a BIM-based workflow connecting envelope design, BIM modelling, environmental analysis and sustainability evaluation.

5.Research Questions

1. How can BIM-based environmental analysis be integrated into building-envelope design to support sustainable decision-making?
2. How effectively can BIM-based tools evaluate building-envelope performance?
3. Which façade and building-envelope parameters have the greatest influence on building environmental performance?
4. How can the findings be integrated into a BIM-based workflow for sustainable façade and building-envelope design?

6.Scope & Limitations

The study will focus on the building envelope as a component of sustainable architectural design and its evaluation through BIM-based workflows.

Limitations:

1. Focus primarily on the building envelope.
1. The study Concentrates on selected environmental-performance parameters.
2. Evaluate selected BIM-based analysis tools rather than all available software.
3. Develop a practical workflow rather than a complete building-performance simulation methodology

7.Significance of study

The research is significant because it addresses the gap between architectural envelope design and environmental performance analysis. The proposed workflow can help architects understand when and how environmental analysis can be incorporated into façade and envelope decisions. The study can identify envelope strategies that contribute to reduced energy demand, improved daylight and improved thermal performance.

8.Research Methodology

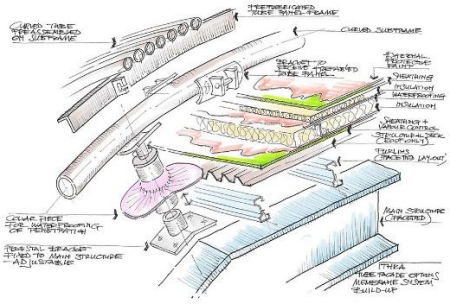
The research will adopt a mixed-method, case-study-based methodology combining qualitative and quantitative approaches.

Literature Review → Case Study Analysis → BIM-Based Façade Analysis → Comparative Evaluation → Development of Integrated BIM Workflow

9.Case Studies

1.King Abdulaziz Centre for World Culture (ITHRA), Saudi Arabia





Project Overview:

- Location: Dhahran, Saudi Arabia
- Architect: Snohetta
- Completion: 2017
- Building Area: 80,000+ m²
- Major Functions: Library, Museum, Theatre, Cinema, Great Hall, Knowledge tower, Energy exhibit.
- Façade materials: Stainless steel tubes
- Climate: Hot, Arid/ Semi-Arid with extremely high summer temperature.
- Sustainability Certification: LEED Gold O+M; LEED Platinum O+M (2026)

I. <u>ENVELOPE DESIGN:</u>
   - A double envelope consisting of an insulated, weather tight building envelope wrapped by an external stainless steel tube skin.
   - Envelope Strategy: The external steel tubes act as a secondary shading layer, reducing direct solar radiation before it reaches the primary envelope. A ventilated cavity removes accumulated heat.
   - Orientation: Interconnected pebble like volumes, façade response is largely achieved through continuous external shading veil and controlled openings.
   - Glazing systems: Concentrated at select openings, Integrated behind steel tube skin. Tubes flatten around window area.
   - Shading devices: Stainless steel tubes create external shading system.
   - Façade materiality: Reflective Stainless-steel tubes.
   - Thermal Insulation: Weather Tight Envelope, buffers external heat and humidity.

II. <u>ENVIRONMENTAL RESPONSE</u>

- Solar Control: Steel tubes reflect and intercept radiation before it reaches main envelope. Skin reduces direct solar-exposure, cavity allows heat to dissipate.
- Daylight Optimization: Steel tubes become flat near windows to allow controlled daylight penetration. Large windows, solar protection.
- Natural Ventilation: The system uses airflow to reduce heat accumulation at the façade. Relies on artificial ventilation.
- Thermal comfort: Supported through External solar shading, Insulated envelope, façade air cavity, Controlled glazing, HVAC & Building service design.

III. PERFORMANCE

- Energy Efficiency: The façade reduces the cooling burden by limiting solar heat entering the building.
- Solar radiation: The external stainless-steel tubes protect the building from solar radiation. Rather than allowing the primary façade to receive direct solar exposure, radiation is intercepted by the external layer.
- Peak Thermal Loads: The cooling system has a reduced external heat load compared to highly glazed façade.

IV. INNOVATION: Transformation of a traditional element into a digitally fabricated façade. The secondary skin provides solar shading, heat dissipation, and architectural identity.

2.Al Faisaliah Tower, Saudi Arabia



Project Overview:

- Location: Riyadh, Saudi Arabia
- Architect: Foster+Partners
- Completion: 2000
- Building Height: 267m, 30 floors
- Major Functions: Offices, hotels, retail, conference, restaurants.
- Façade materials: Glazed curtain wall, aluminium sun shades.

- Climate: Hot, Arid/ Desert.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:
   - Layered glazed curtain wall façade, with Aluminium sun shading.
   - Envelope Strategy: Controlled glazing combined with external shading to reduce glare and solar exposure.
   - Orientation: The envelope is treated as a continuous response to climate in Riyadh.
   - Glazing systems: Non reflective, energy efficient glazing used as external shading system. The tower also incorporates tempered glass elements.
   - Shading devices: Cantilevered Horizontal aluminium sun-shades.
   - Façade materiality: Aluminium panels, glazed curtain wall.
   - Thermal Insulation: Insulation is incorporated within the building envelope.

II. ENVIRONMENTAL RESPONSE
   - Solar Control: The cantilevered panels act as the primary solar-control mechanism.
   - Daylight Optimization: The Tower uses non reflective glass in combination with external shading. This allows for daylighting while keeping out glare.
   - Natural Ventilation: The Building relies on artificial ventilation.
   - Thermal comfort: Supported through External solar shading, Controlled glazing, reduced glare.

III. PERFORMANCE
   - Energy Efficiency: The tower uses comparatively less electricity than similar sized complexes in Saudi Arabia.
   - Solar radiation: The sun-shades prevents a significant amount of direct solar radiation from reaching the glazing.
   - Peak Thermal Loads: The cooling system has a reduced external heat load.

IV. INNOVATION: Integration of environmental shading with the towers structural system.

3.Al Majdoul Tower, Saudi Arabia



Project Overview:

- Location: Riyadh, Saudi Arabia
- Architect: Zeidler Partnership Architects
- Completion: 2020
- Building Height: 244m, 54 floors
- Major Functions: Offices, Residential.
- Façade materials: Glazed curtain wall, aluminium sun shades.
- Climate: Hot, Arid/ Desert.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Glazed High-rise envelope wrapped around a continuous twisted tower.
- Envelope Strategy: Controlled Glazed façade, combined with the towers rotating floor plates.
- Orientation: The Twisting form continuously changes the orientation of the façade. This action distributes the solar exposure across different sides.
- Glazing systems: Glazed curtain wall treatment.
- Shading devices: Horizontal façade elements, self-shading effect by the twisting form.
- Façade materiality: Glazed curtain wall.
- Thermal Insulation: Insulation is a part of the conditioned high rise envelope.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The tower's continuously rotating form changes the angle at which sunlight strikes the facade. This provides self-shade.

- Daylight Optimization: The extensive glazed facade provides daylight and external views. This also means that different parts of the tower will achieve daylighting differently.
- Natural Ventilation: The Building relies on artificial ventilation.
- Thermal comfort: Supported through extensive HVAC systems integrated with centralised controls.

III. PERFORMANCE

- Energy Efficiency: The project's intelligent controls and integration are important aspects which led to it achieving an efficient indoor climate.
- Solar radiation: The twisted geometry modifies the solar exposure of each facade as the tower rises.
- Peak Thermal Loads: The building has an extensive HVAC and controls infrastructure designed for the Riyadh climate.

IV. INNOVATION: A continuously twisting envelope that changes facade orientation with height, while creating a dynamic solar response, forming the tower's architectural identity.

4.Jeddah Tower, Saudi Arabia



Project Overview:

- Location: Jeddah, Saudi Arabia
- Architect: Adrian Smith + Gordon Gill Architecture.
- Completion: Expected by 2030
- Building Area: Approx. 530,000 m²
- Building Height: 1000+m
- Major Functions: Hotels, apartments.
- Façade materials: high-performance glazing, aluminium panels.
- Climate: Hot, Arid/ Desert.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- High-performance glazed curtain-wall envelope integrated with a tapered, three-wing Y-shaped tower.
- Envelope Strategy: Aerodynamic tapered form, controlled high-performance glazing, recessed façade and balconies.
- Orientation: Three-wing Y-shaped plan; one wing is oriented toward Mecca and another, points approximately north, helping reduce solar exposure.
- Glazing systems: High-performance insulated curtain-wall glazing.
- Shading devices: Recessed balconies, projecting floor plates and the deep triangular spaces between the tower's wings create self-shading.
- Façade materiality: High-performance glass, aluminium curtain walls.
- Thermal Insulation: High-performance insulated curtain wall with thermally controlled glazing.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The Tower uses several layers of solar protection, Y Shaped geometry, recessed façade, high controlled glazing.
- Daylight Optimization: The tower retains extensive glazing to provide daylighting, and views. At the same time the solar control glazing limits the excessive solar transmission.
- Natural Ventilation: The system relies on artificial ventilation.
- Thermal comfort: The façade is designed to limit external heat gain before mechanical systems feel the need to compensate.

III. PERFORMANCE

- Energy Efficiency: Jeddah Tower was designed as a high-performance, energy-conscious and sustainability driven project.
- Solar radiation: The combination of the façade envelope reduces the amount of direct solar radiation reaching the interiors.
- Peak Thermal Loads: The envelope is designed to reduce the solar component of cooling loads through solar-control glazing and shading.

IV. INNOVATION: The building's aerodynamic Y-shaped geometry simultaneously acts as the architectural form, structural system and environmental shading mechanism.

5.M9-C, France



Project Overview:

- Location: Paris, France
- Architect: BP Architectures.
- Completion: Expected by 2012
- Building Area: 17,500 m²
- Building Height: Approx. 25m
- Major Functions: Residential, Educational, cultural.
- Façade materials: Aluminium shutters, brick shingles.
- Climate: Temperate.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Layered facade combining masonry/cladding with balconies and operable external aluminium shutters.
- Envelope Strategy: Kinetic external shading layer that can be opened, folded or closed according to sunlight and user preference.
- Orientation: L-shaped building positioned along the northwest and northeast edges of the site; the internal facades face south and receive stronger solar exposure.
- Glazing systems: Residential windows positioned behind the external shutter system.
- Shading devices: Perforated folding aluminium shutters along balconies.
- Façade materiality: Varnished brick shingles, Buxy stone, aluminium shutters and concrete.
- Thermal Insulation: The apartments comply with applicable French thermal standards.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The internal facades face south and use more extensive shutter coverage. The perforated shutters then block excessive direct solar radiation and reduce heat gain.
- Daylight Optimization: The shutters are perforated rather than opaque, which ensures that daylight is penetrated into the building.
- Natural Ventilation: The operable shutters allow residents to modify the facade condition.
- Thermal comfort: The project incorporates several passive comfort measures such as, external solar protection, insulated residential envelope, and operable façade elements.

III. PERFORMANCE
- Energy Efficiency: The environmental approach combines building orientation, solar shading, daylighting and mixed-use planning.
- Solar radiation: The perforated shutters provide an adjustable external solar-control layer. The strategy is particularly effective on the south-facing internal facades, where the project provides more shutters to address increased solar exposure.
- Peak Thermal Loads: The external shutters reduce direct solar heat gain, which should reduce cooling demand during periods of high solar exposure.

IV. INNOVATION: The key innovation is the operable perforated aluminium shutter system.

6.Octagon, UK



Project Overview:

- Location: Birmingham, UK
- Architect: Howells
- Completion: 2025
- No. of Units: 370
- Building Height: 155m, 49 floors

- Major Functions: Residential
- Façade materials: Precast concrete panels, glazing
- Climate: Temperate
- Sustainability Certification: Nil

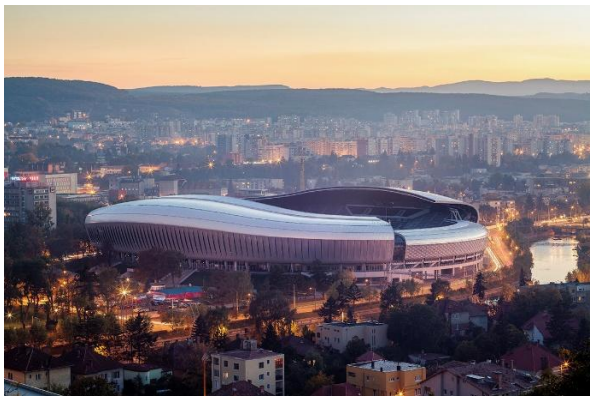
I. ENVELOPE DESIGN:
- Eight-sided high-rise envelope using prefabricated precast-concrete facade panels with ribbon windows.
- Envelope Strategy: Efficient octagonal geometry combined with controlled glazing, and thermal façade.
- Orientation: Eight equal facade faces provide multiple orientations and relatively balanced daylight exposure.
- Glazing systems: High-performance thermal and acoustic glazing forming horizontal window bands.
- Shading devices: Primarily self-shading; facade design also responds to orientation with controlled glazing.
- Façade materiality: Precast concrete panels with two-tone painted finish and high-performance glazed facade elements.
- Thermal Insulation: High-performance insulated facade and glazing

II. ENVIRONMENTAL RESPONSE
- Solar Control: solar response comes mainly from eight-sided geometry, orientation of each façade, controlled glazing ratios, and self-shading.
- Daylight Optimization: The eight facades provide extensive perimeter frontage to the apartments and help distribute daylight and views around the building.
- Natural Ventilation: The system relies on artificial ventilation.
- Thermal comfort: Thermal comfort is supported through; high-performance glazing, insulated facade construction, solar-control measures, and acoustic facade performance.

III. PERFORMANCE
- Energy Efficiency: The project adopts a passive-design approach, with the facade developed to balance daylight, solar gains, solar protection and views
- Solar radiation: The Octagon does not depend on a single facade element to control solar radiation.
- Peak Thermal Loads: The facade strategy reduces potential solar gains through; controlled glazing, orientation, self-shading, and high performance glazing.

IV. INNOVATION: Using the octagonal geometry to simultaneously reduce facade quantity, distribute daylight and views, respond to solar orientation, improve structural efficiency and enable prefabricated facade construction.

7.Cluj Arena, Romania



Project Overview:

- Location: Cluj-Napoca, Romania
- Architect: Dico & Tiganas
- Completion: 2011
- Building Area: Approx. 43,000 m²
- Major Functions: Football stadium, athletics track.
- Façade materials: Perforated metal sheets, woven metal strips and glazed façade elements.
- Climate: Cold/warm-summer humid climate
- Sustainability Certification: Nil

I. <u>ENVELOPE DESIGN:</u>

- A large-scale stadium envelope formed around four major structural frames, with a lightweight translucent façade wrapping the spectator stands.
- Envelope Strategy: The façade uses a translucent outer skin made from woven strips and perforated metal sheets.
- Orientation: The stadium is integrated into the existing urban promenade and sports corridor along the Someș River.
- Glazing systems: Transparent and translucent façade elements.
- Shading devices: The perforated and woven outer façade acts as a filtered solar layer, controlling direct sunlight while maintaining transparency and views.
- Façade materiality: White perforated sheet and woven metal strips.
- Thermal Insulation: No data available.

II. <u>ENVIRONMENTAL RESPONSE</u>

- Solar Control: The perforated and woven façade filters direct sunlight while allowing light to pass through. The reflective surface responds to changing solar conditions.
- Daylight Optimization: The translucent façade allows diffused daylight into circulation.

- Natural Ventilation: The building allows for natural ventilation.
- Thermal comfort: The envelope provides filtered solar exposure and a translucent external layer.

III. PERFORMANCE
- Energy Efficiency The façade's filtered/translucent character reduces the need for completely opaque enclosure while allowing daylight and limiting direct solar exposure.
- Solar radiation: The perforated sheet and woven strips intercept and filter sunlight.
- Peak Thermal Loads: The external filtered skin can reduce direct solar exposure to enclosed areas.

IV. INNOVATION: The key innovation is the use of a translucent, perforated and woven façade as an environmental and visual interface between the stadium and the city.

8.Museo Soumaya, Mexico



Project Overview:

- Location: Mexico city, Mexico
- Architect: Fernando Romero Enterprise
- Completion: 2011
- Building Area: 17,000 m²
- Building Height: Approx 46m
- Major Functions: Museum, exhibition galleries, auditorium, library, restaurant, gift shop and offices.
- Façade materials: 16,000 hexagonal mirrored-steel/aluminium elements over a steel secondary structure.
- Climate: Temperate highland climate.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Doubly-curved sculptural envelope formed as a rotated rhomboid, supported by a structural system of 28 curved steel columns and seven ring-shaped floor slabs.
- Envelope Strategy: The building uses a continuous double-curved external skin composed of approximately 16,000 individual hexagonal mirrored elements. The outer skin is separated from the primary structure through a secondary space-frame system, creating a layered envelope.
- Orientation: The building's changing geometry modifies the building's appearance and solar exposure depending on the viewer's position and time of day.
- Glazing systems: The museum is largely windowless, with natural light primarily entering through a skylight at the upper level.
- Shading devices: The external hexagonal skin acts as a continuous solar-filtering layer over the primary structure rather than using conventional horizontal or vertical shading devices.
- Façade materiality: Approximately 16,000 mirrored hexagonal steel/aluminium panels create a reflective outer skin inspired by traditional Mexican tiled façades.
- Thermal Insulation: The envelope incorporates a weatherproofing layer behind the secondary structure and external cladding.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The continuous external skin intercepts and reflects solar radiation before it reaches the inner envelope. The mirrored surface also changes its reflectivity depending on sunlight and viewing angle.
- Daylight Optimization: Natural daylight is introduced primarily through the upper sky-light, The envelope limits uncontrolled direct daylight penetration.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Thermal comfort is supported through the layered envelope, external reflective skin and controlled introduction of natural light, combined with building services.

III. PERFORMANCE

- Energy Efficiency: The reflective outer envelope reduces direct solar exposure on the primary building enclosure.
- Solar radiation: The mirrored hexagonal skin forms a continuous external layer that intercepts and reflects solar radiation while reducing direct exposure of the underlying envelope.
- Peak Thermal Loads: The external skin acts as a solar-control layer and can reduce direct solar heat gain to the primary envelope.

IV. INNOVATION: The key innovation is the integration of parametric geometry, structural engineering and digital fabrication to create a doubly-curved envelope.

9.1-Triton square, UK



Project Overview:

- Location: London, UK
- Architect: Arup Associates.
- Completion: 2021 (renovation)
- Building Area: approx. 500,000 sqft
- Building Height: 12 floors
- Major Functions: Commercial office
- Façade materials: glazing
- Climate: Temperate climate
- Sustainability Certification: BREEAM Outstanding,

I. <u>ENVELOPE DESIGN:</u>

- Refurbished façade retaining most of the elements from the previous design.
- Envelope Strategy: The project prioritised retention and reuse of the existing building envelope rather than demolition and replacement.
- Orientation: Rather than redesigning the envelope completely, the project focused on improving the performance and value of the existing building fabric.
- Glazing systems: The building incorporates extensive glazing, including floor-to-ceiling windows in the building and atrium to increase access to natural daylight.
- Shading devices: No major conventional external shading system is identified.
- Façade materiality: The existing glazed façade panels were retained and refurbished. Panels were transported to a temporary factory, where they were inspected, deep-cleaned and refurbished before being returned to the building.
- Thermal Insulation: The extended stair cores were positioned outside the thermal line, improving energy performance and reducing anticipated heating requirements.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Environmental performance was improved through the wider retrofit strategy and optimisation of building systems rather than relying solely on new façade components.
- Daylight Optimization: Extensive glazing and floor-to-ceiling windows provide natural daylight to office and atrium spaces. The transformed atrium extends vertically to a large skylight, bringing daylight deep into the building.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Thermal performance was improved through the refurbishment of the existing envelope, relocation of stair cores outside the thermal line and extensive upgrades to mechanical and environmental systems.

III. PERFORMANCE

- Energy Efficiency: New air-handling units, air-source heat pumps and upgraded mechanical infrastructure contribute to improved energy performance.
- Solar radiation: The façade forms the primary interface with solar exposure, while the building's environmental strategy combines the retained envelope with upgraded mechanical systems to manage internal environmental conditions.
- Peak Thermal Loads: The reduction of thermal impact of the existing stair cores and the upgrading of environmental systems contributed to improved building performance.

IV. INNOVATION: The key innovation is the circular refurbishment of an existing high-rise office envelope rather than demolition and replacement. More than 3,000 m² of façade containing over 25,000 components was carefully removed, refurbished and reinstalled.

10.Broadcasting place, UK



Project Overview:

- Location: Leeds, UK

- Architect: Feilden Clegg Bradley Studios
- Completion: 2009
- Building Area: 110,000 sqft
- Building Height: Approx. 69 m / 23 storeys
- Major Functions: Mixed-use development, university teaching and office spaces.
- Façade materials: Cor-Ten weathering steel rainscreen panels, aluminium framing and solar-control glazing.
- Climate: Temperate climate
- Sustainability Certification: BREEAM Very Good

I. ENVELOPE DESIGN:

- Stepped and irregular building envelope developed as a sculptural interpretation of Yorkshire's geological landscape, with sharp triangular corners, angular projections and a strongly roof profile.
- Envelope Strategy: The façade uses a Cor-Ten steel rainscreen system over an insulated building envelope. The irregular elevations were individually developed rather than using a uniform façade module, allowing glazing and solid panels to respond to environmental conditions on different parts of the building.
- Orientation: The tower's accommodation is primarily organised around an east–west orientation. The sculptural south elevation is largely free of windows, while the east and west elevations contain grouped windows that respond to the building's orientation and environmental requirements.
- Glazing systems: The façade incorporates fixed and opening lights, including fritted Low-E solar-control glazing, fixed transparent glazing, tinted glazing, louvred panels and fixed solid panels.
- Shading devices: shading panels are integrated into the façade as part of the varying glazing-to-solid ratio.
- Façade materiality: Cor-ten steel forms the dominant external rainscreen. The material was selected for its solid, sculptural character and ability to develop a deep red-brown oxidised surface over time
- Thermal Insulation: The façade incorporates insulated construction behind the rainscreen. The use of insulated façade panels and improved window systems contributes to lower heat transfer and improved thermal performance.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The façade was specifically developed to reduce solar penetration and overheating. The design team created its own software to analyse individual façade sections and determine appropriate glazing and shading distributions.
- Daylight Optimization: Computational analysis was used to determine the optimum quantity and location of glazing, allowing high levels of natural daylight while controlling excessive solar gains.

- Natural Ventilation: The building's plans were designed to allow natural ventilation where practicable, although the site's proximity to the inner-city motorway and the building's environmental requirements meant that mechanical ventilation was also an important component of the strategy.
- Thermal comfort: Thermal comfort was addressed through the combination of controlled glazing, shading, façade orientation and solar analysis.

III. PERFORMANCE

- Energy Efficiency: The façade design was developed specifically to reduce cooling loads and energy use through computational analysis of the external elevations.
- Solar radiation: The irregular arrangement of glazing, solid panels and shading elements was computationally derived to control solar radiation across different façade areas. This reduced unnecessary solar penetration while retaining sufficient glazed area for daylight.
- Peak Thermal Loads: Limiting solar penetration through differentiated glazing and shading helped reduce the potential for overheating and associated cooling loads.

IV. INNOVATION: The key innovation is the computationally responsive façade design, where the architects developed project-specific software to analyse individual sections of the building envelope and determine the appropriate distribution of glazing and shading.

11.Sunstar Headquarters, Switzerland



Project Overview:

- Location: Etoy, Switzerland
- Architect: Alhadeff Architects, with SdARCH Trivelli & Associates
- Completion: 2015
- Building Area: 7700m²
- Building Height: 3 storeys
- Major Functions: Corporate headquarters, offices, conference rooms, meeting spaces and informal breakout areas.

- Façade materials: Concrete exterior panels, local Swiss stone, glass, laminated wood walls and wood louvres.
- Climate: Temperate Swiss climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- The building has a three-storey, courtyard-based envelope organized around a central skylit atrium. Its façade is composed as a continuous rhythm of opaque and transparent elements, with the composition varying according to the orientation of each elevation.
- Envelope Strategy: The building uses a layered and orientation-responsive façade, combining opaque insulated wall surfaces with glazed areas and rhythmic wooden louvres
- Orientation: The composition of the façade varies according to the building's orientation.
- Glazing systems: Low-emissivity glazing is used to control heat transfer.
- Shading devices: External movable sunshades and wooden louvres are used to control unwanted solar radiation, particularly during summer. The south-facing wooden louvres help block excessive sunlight.
- Façade materiality: The façade combines concrete panels, glass and wooden louvres, while the ground floor uses local Swiss stone/gneiss.
- Thermal Insulation: All external walls are thermally insulated. The opaque walls consist of laminated wood with exterior insulation, an air curtain and an externally ventilated Fiber C or stone façade.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar energy is controlled through low-emissivity glazing, screen-printed glazing and external movable sunshades.
- Daylight Optimization: The offices receive daylight from both the central courtyard and the narrow building depth. The atrium transmits natural light into the surrounding workspaces.
- Natural Ventilation: The building is designed for natural ventilation. The central atrium acts as an important environmental component, facilitating natural ventilation and recovery of exhaust air.
- Thermal comfort: Thermal comfort is supported through the combination of external thermal insulation, internal thermal mass, controlled solar radiation and natural ventilation.

III. PERFORMANCE

- Energy Efficiency: The energy strategy reduces dependence on artificial lighting and mechanical conditioning through natural daylight, natural ventilation, thermal mass and external insulation.

- Solar radiation: The building plan and fragmentation of the external surface reduce exposure to external radiation. Low-emissivity glazing, screen-printed glazing and external movable sunshades control solar energy during summer.
- Peak Thermal Loads: The use of low-E glazing, external shading, thermal insulation and internal thermal mass reduces unwanted solar heat gain and helps moderate thermal loads.

IV. INNOVATION: The key innovation is the integration of a responsive layered façade with passive environmental strategies.

12.Swisstech Convention, Switzerland



Project Overview:

- Location: Ecublens, Switzerland
- Architect: Richter Dahl Rocha & Associés
- Completion: 2014
- Building Area: 41,822 m²
- Building Height: 15m
- Major Functions: Convention and conference centre.
- Façade materials: Large glazed façades, anodized aluminium roof cladding.
- Climate: Temperate Swiss climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Faceted building envelope formed by a large metallic catamaran-shaped roof that progressively rises from the ground and spans over the glazed public spaces.
- Envelope Strategy: The building combines a large lightweight metallic envelope with extensive glazed façades. The roof is supported by a three-dimensional catamaran-shaped steel structure, while the south and north façades open the major interior spaces towards natural light.
- Orientation: The building has a clear north–south organization, with large glazed surfaces facing north and south. The western façade incorporates the building's principal photovoltaic intervention.

- Glazing systems: Large-scale glazed façades provide daylight to the foyer and auditorium. The south façade creates a strong visual connection with the campus plaza.
- Shading devices: The western façade incorporates approximately 300 m² of translucent dye-sensitized photovoltaic cells, functioning simultaneously as a solar-shading screen and electricity-generating façade.
- Façade materiality: The main roof/external shell is clad in naturally anodized aluminium diamond-shaped tiles, producing the characteristic faceted appearance. This metallic skin contrasts with the extensive transparent glass façades.
- Thermal Insulation: The envelope combines insulated opaque construction with extensive glazing and solar-control elements.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The façade acts as an active solar-control layer. The translucent photovoltaic panels filter direct solar radiation before it reaches the glazed façade, simultaneously producing renewable electricity.
- Daylight Optimization: Natural daylight is a major component of the envelope strategy. Large north- and south-facing glazed surfaces allow daylight to penetrate deep into the foyer and large conference spaces.
- Natural Ventilation: The building relies significantly on mechanical environmental systems.
- Thermal comfort: Thermal comfort is addressed through the combination of solar-control glazing elements, the protective metallic roof, controlled daylight and mechanical heating and cooling.

III. PERFORMANCE

- Energy Efficiency: The building integrates renewable-energy generation directly into its envelope through the photovoltaic façade. The 300 m² western façade generates electricity while simultaneously reducing solar heat gain and cooling requirements.
- Solar radiation: The western photovoltaic façade filters direct solar radiation before it reaches the internal glazed spaces.
- Peak Thermal Loads: The photovoltaic shading layer reduces direct solar heat gain to the glazed western façade, thereby helping to reduce cooling demand and potential peak thermal loads.

IV. INNOVATION: The key innovation is the integration of a multifunctional photovoltaic building envelope.

13.Brandhorst Museum, Germany



Project Overview:

- Location: Munich, Germany
- Architect: Sauerbruch Hutton, Matthias Sauerbruch, Louisa Hutton and Juan Lucas Young.
- Completion: 2008
- Building Area: 12,100 m²
- Building Height: 3 floors
- Major Functions: Museum, exhibition galleries, media and graphic-art galleries.
- Façade materials: Approximately 36,000 vertical ceramic rods, glazed in 23 different colours, positioned in front of a horizontally folded and perforated metal skin.
- Climate: Marine West Coast climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- The museum consists of three interconnected volumes forming an elongated building with a taller volume marking the corner of Munich's museum quarter. The building is wrapped in a distinctive multi-layered, polychromatic façade.
- Envelope Strategy: The envelope is designed as a layered façade system. A horizontally folded and finely perforated metal skin forms the background layer, while ceramic rods are positioned in front of it. The ceramic screen functions as an environmental filter, contributing to solar shading and acoustic protection.
- Orientation: The façade responds to the surrounding urban context rather than using a uniform architectural expression. Its ceramic screen creates varying shadow patterns as sunlight moves across the building during the day.
- Glazing systems: The upper galleries receive light from above, while the ground-floor galleries use an asymmetrically positioned daylight system.
- Shading devices: The 36,000 ceramic rods act as a secondary shading layer, filtering sunlight before it reaches the inner façade.
- Façade materiality: The outer skin consists of 36,000 TERRART®-Baguette ceramic rods, each approximately 4 × 4 × 110 cm, glazed in 23 colours. Behind them is a horizontally folded, finely perforated metal sheet. The perforations also help absorb traffic noise from the surrounding streets.

- Thermal Insulation: The façade layers are positioned in front of the building's substructure and thermal insulation, creating a multi-layered envelope. The façade works together with the building's thermal-mass strategy to maintain stable interior temperatures.

II. ENVIRONMENTAL RESPONSE

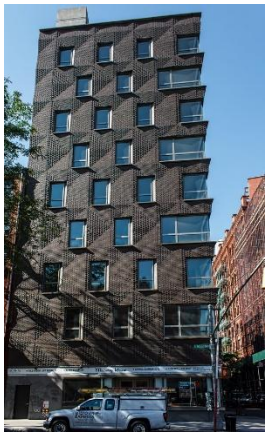
- Solar Control: The external ceramic screen filters solar exposure before it reaches the building envelope. The perforated metal layer and ceramic rods create a secondary façade that reduces direct solar penetration.
- Daylight Optimization: Daylight is a major component of the building's environmental strategy. The lower level receives daylight through a sunken patio, ground-floor galleries receive zenith light through prismatic reflectors, and the upper galleries receive continuous top lighting.
- Natural Ventilation: It uses artificial ventilation.
- Thermal comfort: Thermal stability is achieved through building-component activation, where walls and floors contribute to regulating interior temperatures.

III. PERFORMANCE

- Energy Efficiency: The museum uses an integrated environmental strategy combining building-component activation, groundwater heat pumps, displacement ventilation, controlled daylight and efficient lighting.
- Solar radiation: The layered ceramic façade acts as an external solar-filtering layer. The carefully selected upper-level exterior glazing reduces direct solar radiation.
- Peak Thermal Loads: the combination of the ceramic secondary skin, and controlled glazing, reduces the need for conventional active climate-control systems

IV. INNOVATION: The key innovation is the integration of a highly expressive ceramic secondary façade with environmental performance. The 36,000 individually arranged ceramic rods form a filtering outer skin that provides solar shading, acoustic protection and visual identity while the perforated metal layer behind it further contributes to environmental performance.

14.290 Mulberry house, USA



Project Overview:

- Location: NYC, USA
- Architect: SHoP Architects.
- Completion: 2009
- Building Area: Approximately 26,000–27,000 sq ft.
- Building Height: 13 floors
- Major Functions: Residential condominium building.
- Façade materials: Prefabricated brick curtain-wall panels
- Climate: Humid subtropical climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Undulating brick curtain-wall envelope developed as a contemporary reinterpretation of the historic masonry façades of the surrounding.
- Envelope Strategy: The building uses a rippled, textured brick wrapper around the street-facing elevations.
- Orientation: The two principal street-facing elevations responding to the surrounding masonry context. The north and west façades are particularly important to the envelope strategy.
- Glazing systems: The residences incorporate oversized double-glazed windows, positioned within the rippled masonry façade.
- Shading devices: No major external shading-device system is identified.
- Façade materiality: The principal façade consists of brick embedded in precast-concrete panels.
- Thermal Insulation: The façade incorporates a panelised exterior enclosure and double-glazed windows.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The predominantly masonry envelope significantly limits the amount of exposed glazing on the street façades.

- Daylight Optimization: The use of large double-glazed windows provides natural daylight to the residences while the surrounding brick panels maintain the predominantly masonry character required by zoning.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Thermal comfort is supported by the combination of a predominantly solid masonry envelope and oversized double-glazed windows.

III. PERFORMANCE

- Energy Efficiency: The envelope's principal performance contribution comes from its high proportion of solid masonry, controlled glazing and double-glazed windows.
- Solar radiation: The rippled brick enclosure creates a substantially more solid façade than a conventional curtain-wall building. The projecting brick geometry and recessed glazing help moderate direct solar exposure.
- Peak Thermal Loads: The combination of solid masonry, limited glazing and double-glazed windows can reduce direct solar heat gain and conductive heat transfer compared with an extensively glazed envelope.

IV. INNOVATION: The key innovation is the development of a panelised, digitally fabricated brick curtain wall that transforms conventional masonry into a complex, undulating building envelope.

15.CCTV Headquarters, China



Project Overview:

- Location: Beijing, China
- Architect: OMA, Rem Koolhaas and Ole Scheeren
- Completion: 2012
- Building Area: Approximately 473,000 m²
- Building Height: Tower 1 – 234 m, 54 floors; Tower 2 – 210 m, 44 floors.

- Major Functions: Television studios, administration, offices, news production, broadcasting and programme production facilities.
- Façade materials: Double-glazed curtain wall with low-emissivity glass and an external structural steel diagrid consisting of columns, beams and diagonal bracing members.
- Climate: monsoon-influenced humid climate.
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- The CCTV Headquarters is conceived as a continuous three-dimensional loop rather than a conventional vertical skyscraper. The entire form is wrapped by an irregular structural diagrid façade, making the building envelope simultaneously function as the primary structural system.
- Envelope Strategy: The building uses a diagrid envelope, in which the external steel structure carries significant vertical and lateral loads while forming the visible façade pattern. Behind the structural grid is a glazed curtain-wall system.
- Orientation: The two towers are deliberately inclined toward one another and connected at different orientations to create the continuous loop.
- Glazing systems: The envelope incorporates a glass curtain-wall system, with sources describing the façade as using double-glazed, low-emissivity glass.
- Shading devices: No shading devices identified.
- Façade materiality: The principal visible elements are the steel diagrid and glazed curtain wall. The irregular diagonal steel members form a continuously changing pattern across the façade.
- Thermal Insulation: The curtain-wall system uses double glazing with Low-E coating, providing improved thermal insulation compared with conventional single glazing.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The extensive glazed envelope is combined with Low-E double glazing, which reduces heat transfer and solar heat gain compared with conventional glazing. The deep external diagrid also creates areas of self-shading across portions of the glazed façade.
- Daylight Optimization: The extensive glazed curtain wall provides natural daylight to the office and production areas. The continuous loop and large central void create varied façade orientations and allow daylight to enter spaces along the perimeter.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Thermal comfort is supported primarily through the double-glazed Low-E curtain wall and mechanical environmental systems. The envelope reduces heat transfer through the glazed façade, while the building's large structural mass contributes to thermal stability.

III. PERFORMANCE

- Energy Efficiency: The use of double-glazed Low-E curtain-wall units improves the thermal performance of the extensive glazed façade by reducing heat transfer and solar heat gain.
- Solar radiation: Solar radiation is controlled primarily through the Low-E double-glazed curtain wall and the geometry of the external structural diagrid.
- Peak Thermal Loads: The use of Low-E double glazing and the partially self-shading diagrid helps limit solar heat gain, potentially reducing cooling demand.

IV. INNOVATION: The key innovation is the integration of the building envelope with the primary structural system through the external diagrid.

16.Foundation Louis Vuitton, France



Project Overview:

- Location: Paris, France
- Architect: Frank Gehry
- Completion: 2014
- Building Area: Approx. 7,000 m²
- Building Height: Approx. 46m
- Major Functions: Contemporary art museum and cultural centre.
- Façade materials: High-performance fibre-reinforced concrete panels
- Climate: Temperate oceanic climate
- Sustainability Certification: The project achieved HQE (Haute Qualité Environnementale)

I. ENVELOPE DESIGN:

- Complex layered envelope composed of solid white volumes enclosed and surrounded by twelve large, independently curved glass sails, creating a transparent outer skin around the opaque building mass.

- Envelope Strategy: The Foundation uses a double-layered architectural envelope. The inner building consists of irregular white Ductal-clad volumes, while an independent system of twelve large glass sails surrounds the building.
- Orientation: The glass sails are individually positioned and oriented around the building rather than following a conventional flat façade.
- Glazing systems: The twelve sails comprise approximately 3,600 individually curved laminated glass panels, each produced with its own geometry.
- Shading devices: Solar control is integrated into the glazing itself. The glass sails use fritted glass, with screen-printing and fritting used to control daylight penetration, reduce solar radiation and heat gain.
- Façade materiality: The opaque volumes are clad in approximately 19,000 white Ductal fibre-reinforced concrete panels, many with individually varying curvature. The outer sails use curved glass supported by wooden beams and stainless-steel structures
- Thermal Insulation: The complex glazed façades were required to achieve high standards of air/water tightness, thermal insulation and fire resistance despite their highly irregular geometry.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar radiation is controlled through the fritted and screen-printed glass of the sails. The frit reduces the amount of solar radiation and daylight entering through the glazed envelope, helping to limit solar heat gain.
- Daylight Optimization: The extensive glass envelope provides significant natural daylight while maintaining visual connections with the surrounding. Fritting and screen-printing regulate the intensity of daylight entering the building, preventing the large areas of glazing from creating uncontrolled light penetration.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Supported through the combination of the insulated opaque volumes, solar-controlled glazing and the layered relationship between the glass sails and inner building.

III. PERFORMANCE

- Energy Efficiency: The building incorporates several environmental strategies beyond the façade. The glass roof/sail system allows rainwater collection and reuse.
- Solar radiation: The fritted glass sails reduce solar radiation and control the amount of daylight entering the building. The layered envelope creates a buffer between the external environment and the inner exhibition volumes, while the changing geometry of the sails modifies solar exposure throughout the building.
- Peak Thermal Loads: The combination of fritted glazing, screen-printing, the double-layered envelope and largely opaque inner volumes helps limit

excessive solar heat gain. The reduction in direct solar radiation reduces the potential for cooling demand associated with the extensive glazed envelope.

IV. INNOVATION: The key innovation is the integration of advanced digital modelling, computational geometry and digital fabrication to construct an extremely complex, non-repetitive building envelope.

17.Louvre Museum, UAE



Project Overview:

- Location: Abu Dhabi, UAE
- Architect: Ateliers Jean Nouvel
- Completion: 2017
- Building Area: Approximately 26,000 m² of gross floor area.
- Building Height: The dome reaches approximately 36 m above ground level, with a diameter of approximately 180 m.
- Major Functions: Universal museum
- Façade materials: The museum buildings are formed with approximately 3,900 ultra-high-performance fibre-concrete (UHPC) panels, while the enormous dome consists of eight layers of stainless-steel and aluminium cladding over a steel structural system.
- Climate: Hot desert climate
- Sustainability Certification: LEED Silver and achieved a 3 Pearl Estidama Design Rating.

I. ENVELOPE DESIGN:

- Large-scale floating perforated dome forming a multilayered climatic envelope over a low-rise "museum city" inspired by the traditional Arabian medina.
- Envelope Strategy: The project uses a layered environmental envelope in which 55 low-rise pavilion buildings are placed beneath a vast 180 m diameter dome. The dome acts as a large-scale external shading device, creating a protected microclimate over the buildings

- Orientation: The museum is organised as a cluster of low-rise buildings surrounded by water, with the dome covering approximately two-thirds of the complex. Rather than responding through conventional individual façades, the environmental strategy operates at the scale of the entire museum-city, with the dome controlling solar exposure across the spaces below.
- Glazing systems: The individual museum buildings use controlled glazed openings within their UHPC envelopes. The galleries are enclosed and highly controlled to protect artworks, while the external circulation areas remain more open beneath the dome.
- Shading devices: The 180 m diameter dome is the principal shading device.
- Façade materiality: The pavilion buildings use approximately 3,900 UHPC panels, The dome contains four outer stainless-steel layers and four inner aluminium layers, separated by a substantial structural frame.
- Thermal Insulation: The enclosed museum buildings incorporate an airtight and insulated envelope to separate the tightly controlled gallery environment from Abu Dhabi's extreme climate.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar control is the primary environmental function of the dome. Its eight overlapping layers progressively filter intense Abu Dhabi sunlight before it reaches the museum buildings and outdoor circulation areas.
- Daylight Optimization: The dome does not simply block daylight; it modulates and diffuses it. Environmental modelling established the required transparency and perforation ratios for the eight cladding layers.
- Natural Ventilation: The dome's multilayered construction creates a large shaded external environment, while the spaces between the dome layers contribute to the overall environmental strategy. However, the enclosed museum galleries rely primarily on controlled mechanical systems.
- Thermal comfort: The dome creates a microclimate beneath the building envelope, substantially reducing direct solar exposure to the courtyards and pedestrian routes. The combination of shading, reflective materials, thermal mass, airtight insulated envelopes and efficient HVAC systems maintains comfortable internal conditions.

III. PERFORMANCE

- Energy Efficiency: The dome significantly reduces the amount of solar energy reaching the museum buildings, thereby lowering the cooling requirement. The project combines this passive strategy with efficient HVAC, lighting and sanitary systems.
- Solar radiation: The dome functions as a large-scale solar filter. Environmental analysis determined the required transparency of the different dome layers so that sunlight would be sufficiently reduced while retaining the desired visual and daylight effect.

- Peak Thermal Loads: By shading the museum buildings and external circulation spaces before solar radiation reaches them, the dome reduces the solar heat gain imposed on the conditioned spaces. The resulting lower heat gain helps reduce cooling demand and potential peak cooling loads.

IV. INNOVATION: The key innovation is the creation of a building-scale environmental envelope in which the dome operates simultaneously as architecture, structure, solar-control system and climatic moderator.

18.Heydar Alivey Centre, Azerbaijan



Project Overview:

- Location: Baku, Azerbaijan
- Architect: ZHA
- Completion: 2013
- Building Area: Approx. 101,801 m²
- Building Height: 74m
- Major Functions: Cultural centre comprising exhibition spaces, museum/gallery areas, auditorium and multipurpose cultural facilities.
- Façade materials: Glass Fibre Reinforced Concrete (GFRC), Glass Fibre Reinforced Polyester (GFRP), glass and steel.
- Climate: Semi-arid climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- The Heydar Aliyev Centre is defined by a continuous, fluid envelope that rises from the surrounding plaza and folds over the building to form its roof and walls. Rather than separating the building, façade and landscape, the design creates a seamless transition between ground, exterior envelope and interior.
- Envelope Strategy: The envelope was developed as a free-form, continuous skin capable of accommodating different structural, functional and technical requirements. A rigid space-frame forms the underlying structure, while the external cladding follows the continuously changing geometry through rationalised panel seams.

- Orientation: The fluid surface rises from the plaza, folds over the building and creates different orientations across the continuously changing roof and façade surfaces.
- Glazing systems: Semi-reflective glass is incorporated within the white envelope. The glazing provides controlled visual connections between interior and exterior while maintaining the overall continuity of the fluid surface.
- Shading devices: No conventional external horizontal or vertical shading system is identified in the supplied sources. Solar exposure is instead addressed through the geometry, curvature and continuous layering of the envelope.
- Façade materiality: GFRC and GFRP panels were selected as the principal free-form cladding materials because they could accommodate the complex plasticity of the building.
- Thermal Insulation: The envelope forms a continuous enclosure, forming a closed habitat for the internal environment. The interiors are mechanically ventilated.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The continuous curved envelope limits the use of conventional façade shading devices by using surface geometry and controlled glazing as part of the environmental response. The semi-reflective glass also reduces visual transparency.
- Daylight Optimization: Daylight enters through carefully positioned glazed areas and openings within the fluid envelope. The use of semi-reflective glass provides controlled daylight and views while maintaining the visual continuity of the exterior surface.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: The complex envelope provides a continuous enclosure around the large interior spaces, while the building's structural and mechanical systems contribute to maintaining controlled internal conditions.

III. PERFORMANCE

- Energy Efficiency: The continuous envelope, controlled glazing and integration of structural and enclosure systems contribute to the building's environmental response.
- Solar radiation: The building's highly curved envelope produces continuously changing surface orientations, causing solar radiation to interact differently with the façade throughout the day. The use of semi-reflective glazing and the deep, continuously folded surface helps control direct solar exposure.
- Peak Thermal Loads: The relatively limited glazed area, predominantly opaque external skin and controlled semi-reflective glazing can reduce direct solar heat gain compared with a fully glazed façade.

IV. INNOVATION: The key innovation is the integration of parametric geometry, structural engineering, digital modelling and free-form façade fabrication to create a continuous building envelope.

19.Vancouver house,



Project Overview:

- Location: Vancouver, British Columbia, Canada
- Architect: Bjarke Ingels Group (BIG)
- Completion: 2020
- Building Area: Approximately 60,600 m²
- Building Height: 157m, 52 residential floors
- Major Functions: Mixed-use development comprising residential apartments, rental housing, retail, commercial spaces, restaurants.
- Façade materials: High-performance glazed curtain wall, aluminium façade panels, framed balconies, and substantial concrete structural elements.
- Climate: Temperate oceanic climate
- Sustainability Certification: LEED Platinum

I. ENVELOPE DESIGN:

- Dramatically expanding triangular-to-rectangular tower envelope shaped by the constraints of the Granville Street Bridge, surrounding views, solar conditions and neighbouring public spaces.
- Envelope Strategy: The building begins with an extremely narrow triangular footprint at its base and progressively expands outward as it rises, eventually becoming a substantially larger rectangular floor plate.
- Orientation: The four façades have different environmental and geometric responses. The north and east façades respond primarily to the incremental

cantilever and changing floor geometry, while the south and west façades respond more directly to solar exposure.

- Glazing systems: Extensive glazed curtain-wall areas provide views, daylight and natural connections to Vancouver's surrounding landscape. The glazing is integrated with projecting balconies and recessed residential spaces, creating variation in depth across the façade.
- Shading devices: South-facing loggias and west-facing balconies are enclosed by thick projecting frames, providing passive solar shading and privacy.
- Façade materiality: The envelope combines glass curtain wall and aluminium panels with the building's concrete structural system.
- Thermal Insulation: The high-performance glazed envelope and insulated opaque façade components contribute to the building's energy performance.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar response is directly incorporated into the building's geometry. The south-facing loggias and west-facing balconies use deep projecting frames to provide passive shading, reducing direct solar exposure to the glazed areas.
- Daylight Optimization: The expanding form and extensive glazing provide residents with access to daylight and views. BIG's environmental approach considered the relationship between the tower's geometry, neighbouring buildings and the preservation of daylight and views within the surrounding urban environment.
- Natural Ventilation: Natural ventilation is incorporated into the broader sustainable-design strategy, although Vancouver House remains a mechanically conditioned high-rise residential building.
- Thermal comfort: Thermal comfort is supported through the combination of high-performance glazing, orientation-specific shading, recessed balconies and energy-efficient building systems.

III. PERFORMANCE

- Energy Efficiency: Vancouver House's LEED Platinum rating reflects an integrated sustainability strategy combining high-performance building systems, efficient envelope design, sustainable transportation, daylighting and passive environmental measures.
- Solar radiation: Solar exposure was a direct determinant of the façade configuration. The south and west façades incorporate deeper loggias and balcony frames, which intercept direct sunlight before it reaches the glazed envelope. The different treatment of each façade demonstrates orientation-responsive envelope design.
- Peak Thermal Loads: Passive shading through the deep south-facing loggias and west-facing balcony frames reduces direct solar heat gain to residential spaces. This can reduce cooling requirements and associated peak loads.

IV. INNOVATION: The key innovation is the environmentally and structurally responsive transformation of the tower from a tiny triangular footprint into a large rectangular floor plate. The unusual geometry was generated from the site's constraints rather than imposed as an arbitrary sculptural form.

20.KAFD Metro station, Saudi Arabia



Project Overview:

- Location: Riyadh, Saudi Arabia
- Architect: ZHA
- Completion: 2024
- Building Area: Approx. 45,000 m²
- Building Height: Approx. 32m
- Major Functions: Metro interchange, six rail platforms, passenger circulation, public spaces, shops, parking, prayer rooms.
- Façade materials: Ultra-high-performance concrete (UHPC) panels with steel supporting structure.
- Climate: Hot-arid desert climate
- Sustainability Certification: LEED Gold certification

I. ENVELOPE DESIGN:

- The station is enclosed by a continuous, highly articulated external envelope derived from a series of opposing sine-wave geometries. The envelope acts simultaneously as an architectural skin, solar-control device and self-supporting structural system.
- Envelope Strategy: The envelope combines passive solar protection, structural efficiency and environmental performance within a single external system. The Geometric perforations reduce solar gain.
- Orientation: The envelope responds primarily to the station's circulation network and climatic conditions rather than following a conventional directional façade strategy. Its continuously changing geometry creates varying surface orientations and self-shading conditions.

- Glazing systems: Glazing is incorporated behind the external perforated envelope to provide controlled daylight and visual connection while the outer skin limits direct solar exposure.
- Shading devices: The perforated UHPC external skin acts as the principal shading system. Its geometric openings and varying density reduce direct solar exposure and solar heat gain.
- Façade materiality: The external skin uses ultra-high-performance concrete panels supported by a substantial steel structure.
- Thermal Insulation: The building envelope uses a high-performance UHPC external skin combined with insulated façade construction to limit heat transfer from Riyadh's extreme external temperatures.

II. ENVIRONMENTAL RESPONSE
- Solar Control: Solar control is primarily achieved through the perforated external UHPC envelope. The openings reduce direct solar exposure while maintaining a filtered relationship between the exterior and interior.
- Daylight Optimization: The perforated façade allows filtered natural light to enter the station while reducing excessive direct solar penetration. The combination of openings, large internal volumes and the continuous envelope helps create a controlled daylight environment.
- Natural Ventilation: The building relies on artificial ventilation.
- Thermal comfort: Thermal comfort is achieved through the combination of passive shading, reduced solar gain, high-efficiency cooling and controlled air retention.

III. PERFORMANCE
- Energy Efficiency: The station combines passive envelope strategies with a high-efficiency cooling system powered by renewable energy.
- Solar radiation: The façade's geometric perforations are specifically designed to reduce solar gain. The continuously varying external geometry also creates self-shading conditions across portions of the envelope.
- Peak Thermal Loads: The reduction of direct solar radiation through the perforated envelope helps lower the cooling demand associated with the extreme Riyadh climate.

IV. INNOVATION: The major innovation of KAFD Metro Station lies in the integration of parametric envelope geometry, structural engineering and environmental performance into one system.

21. Leeza Soho, China



Project Overview:

- Location: Beijing, China
- Architect: ZHA
- Completion: 2019
- Building Area: Approx. 172,800 m²
- Building Height: 207m, 45 floors
- Major Functions: Grade-A office tower providing flexible office spaces for small and medium-sized businesses, with retail, public spaces.
- Façade materials: Double-insulated, unitised low-E glass curtain wall, with stepped glazing units.
- Climate: Monsoon type humid climate
- Sustainability Certification: LEED Gold certification

I. <u>ENVELOPE DESIGN:</u>

- Dramatically twisting double-volume tower envelope formed around a central atrium, with two building halves enclosed within a single continuous glazed façade shell.
- Envelope Strategy: The tower is divided into two halves because an underground subway tunnel diagonally crosses the site. The two halves are enclosed by a single continuous façade shell, while the space between them forms the full-height central atrium.
- Orientation: The tower's geometry responds directly to the diagonal subway tunnel below the site and the orientation of Lize Road. The atrium rotates through the height of the building, creating changing orientations and views while maintaining a unified external envelope.
- Glazing systems: The façade uses a double-insulated, unitised glass curtain-wall system. Individual glazing units step at an angle on each floor.
- Shading devices: The building uses its own geometry as a shading mechanism. The two halves of the tower shade the central atrium and its public spaces. The low-E glazing further reduces solar heat gain, while the changing angle of the façade provides different degrees of solar exposure across the tower.

- Façade materiality: The external envelope is predominantly a highly transparent unitised glass curtain wall, allowing the twisting atrium to remain visually connected to the city.
- Thermal Insulation: The building uses double-insulated low-E glazing, with an overall external envelope U-value of approximately 0.55 W/m²K.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar control is achieved through the combination of low-E glazing, the self-shading geometry of the two tower halves and the recessed/angled relationship between the atrium and façade.
- Daylight Optimization: The 194.15 m-high atrium acts as a daylight shaft, allowing natural light to penetrate deep into the centre of the tower.
- Natural Ventilation: The atrium operates as a thermal chimney, with an integrated ventilation system. The unitised façade incorporates narrow ventilating registers that can draw outside air through operable cavities when required.
- Thermal comfort: Thermal comfort is supported through the double-insulated low-E façade, controlled solar transmission, thermal-chimney atrium and mechanical environmental systems.

III. PERFORMANCE

- Energy Efficiency: Leeza SOHO integrates the façade with a wider 3D BIM-based energy-management system that monitors environmental control and energy efficiency in real time.
- Solar radiation: The low-E glazing and self-shading atrium work together to reduce solar radiation reaching occupied spaces.
- Peak Thermal Loads: The building envelope's 0.55 W/m²K overall U-value, low-E glazing and shading coefficient of 0.4 reduce heat transfer and solar heat gain.

IV. INNOVATION: The key innovation is the integration of the twisting atrium, structural geometry, façade system and environmental performance into a single building-envelope strategy.

22.Battersea Power-station, UK



Project Overview:

- Location: London, UK
- Architect: Sir Giles Gilbert Scott, WilkinsonEyre
- Completion: 1933–1955; redevelopment completed in 2022
- Building Area: Approx. 120,000 m²
- Building Height: Approx. 50 m to the main roof
- Major Functions: Retail, offices, residential apartments, restaurants
- Façade materials: Original brick envelope, restored and supplemented with approximately 1.75 million bespoke bricks; steel, glass and contemporary façade insertions.
- Climate: Temperate climate
- Sustainability Certification: BREEAM certifications have been recorded for several components of the redevelopment

I. <u>ENVELOPE DESIGN:</u>

- The original massive brick enclosure is retained as the principal architectural envelope. New internal and external interventions are inserted within and around the historic shell, allowing the building to accommodate modern mixed-use functions while preserving its industrial character.
- Envelope Strategy: The redevelopment retains the monumental original brick envelope of the power station while carefully introducing new glazing, residential insertions and contemporary additions.
- Orientation: The building's long industrial form and large turbine halls respond primarily to its original power-generation function. In the redevelopment, the existing envelope was retained, with new openings and interventions carefully positioned to provide daylight, views and access without compromising the listed structure.
- Glazing systems: New windows and glazed insertions were introduced where required for the adaptive reuse. In the residential areas, floor-to-ceiling glazing was incorporated behind the restored brick façade.
- Shading devices: The original massive brick envelope provides substantial solar buffering because of its depth and limited glazing.

- Façade materiality: The primary façade remains predominantly brick, preserving the architectural identity of the original power station.
- Thermal Insulation: New envelope interventions incorporate high-performance insulation systems.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The retained brick envelope provides a substantial thermal and solar buffer, while the relatively controlled amount of glazing limits direct solar gains.
- Daylight Optimization: Daylight was introduced through restored skylights, end windows, the central atrium and new glazed openings.
- Natural Ventilation: The restored turbine halls use a combination of passive environmental characteristics and mechanically assisted ventilation.
- Thermal comfort: The substantial thermal mass of the original structure plays an important role in moderating indoor conditions.

III. PERFORMANCE

- Energy Efficiency: A major part of the environmental strategy is the reuse of the existing building fabric.
- Solar radiation: The thick brick envelope and relatively limited original glazing reduce direct solar exposure to the interior. New glazed areas were introduced selectively, with the overall envelope strategy balancing daylight and solar control.
- Peak Thermal Loads: The thermal mass of the existing structure, controlled glazing and limited solar gain reduce the need for extensive heating and cooling in the large turbine halls.

IV. INNOVATION: The key innovation is the adaptive reuse of a monumental industrial envelope as a high-performance mixed-use building while retaining its historic architectural identity. Rather than replacing the existing brick structure, the project combines heritage conservation with new glazing, insulation, waterproofing, environmental systems and carefully controlled contemporary insertions.

23.Google Bay View Campus, USA



Project Overview:

- Location: California, USA
- Architect: Bjarke Ingels Group, Heatherwick Studio, Adamson Associates
- Completion: 2022
- Building Area: Approx.102,193 m² across the campus
- Building Height: 2 storey building
- Major Functions: Two workplace buildings, a 1,000-person events centre, employee accommodation, collaborative spaces
- Façade materials: Glazing and clerestory windows, opaque canopy construction, tubular steel structure, photovoltaic solar shingles.
- Climate: Warm-summer climate
- Sustainability Certification: LEED Platinum

I. ENVELOPE DESIGN:

- The envelope is conceived as a large environmental canopy rather than a conventional vertical façade, integrating daylighting, solar control, photovoltaic generation and thermal protection into one continuous building skin.
- Envelope Strategy: The campus uses lightweight, large-span catenary canopy structures with clerestory openings and perimeter glazing. The canopy provides a combination of opaque thermal protection and controlled daylight, while the roof itself becomes an active photovoltaic envelope.
- Orientation: The low-rise campus is arranged as three buildings around courtyards and open landscape, with the canopy geometry and clerestories configured to provide widespread daylight and external views while limiting unwanted solar exposure.
- Glazing systems: Continuous clerestory glazing is inserted between the canopy bays to bring daylight deep into the large open-plan workplaces.
- Shading devices: The combination of opaque canopy areas, clerestory positioning and stainless-steel elements helps diffuse direct sunlight and reduce glare.
- Façade materiality: The envelope combines transparent glazing with a highly integrated photovoltaic roof skin. The PV shingles are custom-designed silver-grey, textured/prismatic glass panels that function simultaneously as roofing and energy-generating architectural material.
- Thermal Insulation: The opaque canopy minimizes thermal heat gain and reduces overall energy loads, while the photovoltaic roof forms an additional protective layer over the roof construction.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The opaque portions of the canopy reduce direct solar heat gain, while the PV roof acts as a solar-energy-generating protective skin. The

clerestory openings are carefully controlled to admit daylight without excessive glare.

- Daylight Optimization: Clerestory windows are distributed throughout the canopy to bring daylight deep into the large-span workplace. The strategy provides daylight to most workstations while maintaining views to the outdoors.
- Natural Ventilation: The high-volume canopy spaces accommodate 100% outside-air ventilation, with the large canopy volume supporting the movement of warm air.
- Thermal comfort: Thermal comfort is addressed through the combination of reduced solar heat gain, shaded glazing, high-volume canopy spaces, geothermal heating/cooling and controlled daylight.

III. PERFORMANCE

- Energy Efficiency: ay View is an all-electric campus. Its photovoltaic roof produces approximately 7 MW of renewable energy.
- Solar radiation: The approximately 50,000 photovoltaic panels are arranged as a continuous “dragon-scale” skin across the curved canopy. Their prismatic glass surface captures solar energy while reducing reflected glare, an important consideration because of the nearby active airfield.
- Peak Thermal Loads: The opaque canopy reduces direct thermal heat gain, while clerestories are used selectively rather than exposing the entire roof to glazing.

IV. INNOVATION: The key innovation of Google Bay View is the transformation of the building envelope into an integrated environmental and energy-generating system. Instead of treating photovoltaic panels as equipment added to a conventional roof, the project makes the PV system the architectural roof skin itself.

24.The black & white building, UK



Project Overview:

- Location: London, UK
- Architect: Waugh Thistleton Architects
- Completion: 2022
- Building Area: 4,480 m² Gross Internal Floor Area
- Building Height: 17.8 m
- Major Functions: Flexible commercial office/workspace, including shared workspaces, private offices, meeting spaces, breakout areas, lounges.
- Façade materials: Floor-to-ceiling high-performance glazing, timber/aluminium curtain-wall framing and thermally modified tulipwood external louvres, supported by a secondary timber structure.
- Climate: Temperate climate
- Sustainability Certification: BREEAM Excellent.

I. ENVELOPE DESIGN:

- The façade consists of a floor-to-ceiling glazed curtain wall protected by an external layer of timber louvres. This creates a two-layer envelope in which the inner glazed skin provides weather and thermal enclosure while the outer timber layer performs solar-control functions.
- Envelope Strategy: The project follows an “architecture of sufficiency” approach, where each component is designed to perform a specific function without unnecessary material. The façade combines daylight access, natural ventilation, solar control and low-carbon timber construction.
- Orientation: The louvre configuration changes according to orientation. Horizontal louvres are used on the southern façade, while vertical louvres are used on the east and west façades to respond to different solar angles.
- Glazing systems: The façade uses floor-to-ceiling glazed panels supported by a glulam composite timber/aluminium curtain-wall system.
- Shading devices: The principal shading system is the external thermally modified louvre system. The louvres are positioned at approximately 375 mm centres and increase in depth as they rise up the building
- Façade materiality: The façade combines high-performance glass, timber/aluminium curtain-wall framing and thermally modified solar-shading louvres.
- Thermal Insulation: The SGG COOL-LITE® SKN 183 glazing provides both thermal insulation and solar-control performance while maintaining high light transmission.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The external timber louvres form the primary solar-control layer. Their geometry was developed through parametric modelling of the sun's movement across the façades.

- Daylight Optimization: The façade uses high-light-transmission SGG COOL-LITE® SKN 183 glazing, allowing abundant natural daylight into the workspaces.
- Natural Ventilation: Inward-opening casement windows are incorporated into the floor-to-ceiling glazed façade, allowing the workspaces to be naturally ventilated as part of a mixed-mode ventilation strategy.
- Thermal comfort: The combination of external shading, solar-control glazing and mixed-mode ventilation reduces solar heat gain while maintaining daylight and fresh-air access.

III. PERFORMANCE

- Energy Efficiency: The building achieves an operational energy figure of approximately 44 kWh/m²/year, according to Timber Development UK. The façade contributes through external solar shading, high-performance glazing, daylight optimisation and mixed-mode ventilation.
- Solar radiation: The timber louvres intercept solar radiation before it reaches the glazing, reducing direct solar heat gain while allowing controlled daylight penetration.
- Peak Thermal Loads: External louvres significantly reduce solar gains at the glazed façade, thereby reducing the potential for overheating and associated cooling loads.

IV. INNOVATION: The key innovation is the integration of a mass-timber structure with a timber-based high-performance façade and digitally optimised external shading system.

25.K11 Musea, Hong Kong



Project Overview:

- Location: Hong Kong, China
- Architect: Kohn Pedersen Fox (KPF) & Ronald Lu & Partners (RLP)

- Completion: 2019
- Building Area: 111,483 m²
- Building Height: 10 storeys
- Major Functions: Cultural-retail destination comprising retail spaces, galleries, restaurants, recreation spaces, exhibition areas, cultural facilities.
- Façade materials: Limestone, glass, steel, concrete, aluminium and extensive vegetation. The building also incorporates conical-shaped glass panels at the Sunken Plaza and hand-painted aluminium panels.
- Climate: Humid subtropical climate
- Sustainability Certification: LEED Gold

I. ENVELOPE DESIGN:

- The building envelope is conceived as a stepped artificial landscape, with horizontal terraces and ribbons of vegetation breaking down the scale of the large retail building. The façade combines solid limestone surfaces, glazing and extensive planted areas.
- Envelope Strategy: The envelope combines architectural massing, vegetation and glazing to create a layered environmental response.
- Orientation: The building occupies a prominent waterfront position overlooking Victoria Harbour. The façade was developed with consideration of views, façade performance and window-to-wall ratio, particularly because of the building's exposed harbourfront location.
- Glazing systems: Extensive glazing is incorporated into the retail and public areas, while the Sunken Plaza incorporates a curved glass wall with conical-shaped glass panels.
- Shading devices: The stepped massing, projecting terraces and vegetation provide self-shading and solar-filtering effects across portions of the façade. The extensive greenery also acts as a secondary environmental layer between the external climate and building surfaces.
- Façade materiality: The envelope combines limestone cladding, high-performance glass curtain walls, aluminium/metal panels and extensive integrated vegetation.
- Thermal Insulation: The façade strategy combines controlled glazing, solid limestone-clad areas, planted surfaces and environmental systems to improve the building's overall thermal performance.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The stepped façade, projecting terraces and extensive vegetation help reduce direct solar exposure on portions of the building envelope.
- Daylight Optimization: Natural daylight is a major feature of the internal environmental strategy. The 35 m-high Opera Theatre atrium receives natural

light through a large overhead Oculus, allowing daylight to penetrate deep into the central public space.

- Natural Ventilation: K11 MUSEA primarily relies on mechanically controlled environmental systems rather than conventional façade-driven natural ventilation.
- Thermal comfort: Thermal comfort is supported through the combination of controlled façade design, extensive greenery, and shaded terraces.

III. PERFORMANCE

- Energy Efficiency: The development incorporates a seawater-cooled, oil-free HVAC chiller system, reported to reduce annual energy consumption by more than 12% compared with the U.S. ASHRAE 90.1 baseline.
- Solar radiation: The combination of stepped geometry, projecting terraces, vegetation and controlled façade openings reduces direct solar exposure on parts of the building.
- Peak Thermal Loads: The extensive green façade and stepped terraces provide shading and reduce direct solar exposure, while the efficient seawater-cooled HVAC system addresses the remaining cooling demand.

IV. INNOVATION: The key innovation is the integration of landscape, façade design and environmental performance into a single stepped building envelope. Instead of treating greenery as an independent landscape element, K11 MUSEA incorporates approximately 4,600 m² of living-wall greenery directly into its façade and terraces.

26.Eden, UK



Project Overview:

- Location: Bodelva, UK
- Architect: Nicholas Grimshaw, Grimshaw Architects
- Completion: 2001
- Building Area: Approx. 23,000 m²

- Building Height: 55m
- Major Functions: Botanical biomes, visitor attraction, educational and exhibition spaces, research and environmental interpretation.
- Façade materials: Triple-layer ETFE pneumatic cushions supported by a lightweight tubular steel hex-tri-hex space-frame.
- Climate: Temperate climate
- Sustainability Certification: Nil

I. ENVELOPE DESIGN:

- Geodesic envelope formed by a sequence of eight interconnected dome-like Biomes, developed from the concept of soap bubbles that could adapt to the irregular and unstable geometry of the former clay pit.
- Envelope Strategy: The Biomes use a lightweight two-layer hex-tri-hex steel space-frame carrying inflated ETFE cushions. The structural geometry provides a high enclosed volume with relatively low material use, while the ETFE skin creates a transparent, lightweight environmental enclosure.
- Orientation: Rather than relying on a conventional directional façade, the Biomes follow the contours of the former quarry. Their dome geometry provides a large enclosed volume while allowing the structures to adapt to the site's uneven and shifting ground conditions.
- Glazing systems: Conventional glass is replaced almost entirely by transparent ETFE cushions. Each hexagonal or pentagonal panel consists of three layers of ETFE foil inflated to form cushions approximately two metres deep.
- Shading devices: The envelope does not primarily use conventional external louvers or fins. Solar control is achieved through the properties and configuration of the ETFE cushion system, while the geometry of the Biomes distributes solar exposure across the large transparent surface.
- Façade materiality: The external envelope consists of geometric hexagonal and pentagonal ETFE cushions supported by a tubular steel space-frame. The largest individual hexagonal elements can reach approximately 11 m across.
- Thermal Insulation: The multi-layer inflated ETFE cushions trap air between the membrane layers, creating a lightweight thermal blanket.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The transparent ETFE envelope allows solar radiation to enter the Biomes, which is essential for plant growth and contributes to the creation of the required internal climates.
- Daylight Optimization: Daylight is a fundamental component of the envelope strategy. The highly transparent ETFE cushions allow extensive natural light penetration into the Biomes, reducing the need for artificial lighting during daylight hours and supporting the photosynthesis of the plants.
- Natural Ventilation: Natural lighting and ventilation form part of Eden's broader environmental strategy. The transparent Biomes are climate-controlled

environments, however Building management systems regulate temperatures and building services across the site.

- Thermal comfort: Thermal conditions are controlled according to the individual biome requirements. The layered ETFE cushions provide insulation while the large transparent enclosure permits solar gains necessary for the vegetation.

III. PERFORMANCE

- Energy Efficiency: The envelope minimizes structural and material requirements through its lightweight space-frame and ETFE cushion system.
- Solar radiation: Solar radiation is intentionally admitted through the transparent ETFE skin to support the plants and generate the internal environmental conditions. The envelope therefore works as a solar-gain system rather than attempting to eliminate solar exposure completely.
- Peak Thermal Loads: The three-layer ETFE cushion reduces uncontrolled heat transfer through the envelope while the relatively low mass of the material and large enclosed volume allows the Biomes to respond to changing environmental conditions.

IV. INNOVATION: The key innovation is the integration of geodesic structural geometry, lightweight ETFE pneumatic cushions and environmental performance into a large-scale greenhouse envelope.

27.Bosco Verticale, Italy



Project Overview:

- Location: Milan, Italy
- Architect: Boeri Studio, Stefano Boeri, Gianandrea Barreca and Giovanni La Varra
- Completion: 2014
- Building Area: The two residential towers together contain approximately 36,000 m² of built floor area.
- Building Height: Tower A – approximately 112 m; Tower B – approximately 80 m

- Major Functions: Residential apartments, private terraces and balconies, communal circulation and landscaped outdoor spaces.
- Façade materials: Porcelain stoneware ventilated façade cladding, reinforced concrete, aluminium substructure, stainless-steel brackets, high-performance glazing and extensive vegetation.
- Climate: Humid subtropical climate
- Sustainability Certification: LEED Gold certification

I. ENVELOPE DESIGN:

- The envelope is conceived as a three-dimensional planted façade rather than a conventional flat building skin. Cantilevered terraces extend outward from the residential floor plates and accommodate large planters, creating a vegetated layer in front of the primary building envelope.
- Envelope Strategy: The project replaces the idea of a conventional mineral façade with a living second skin. Vegetation provides shading, filtration and seasonal changes while simultaneously creating private outdoor spaces for residents.
- Orientation: Plant species and their locations were determined according to façade orientation, solar exposure, height, wind and seasonal conditions. South-facing elevations incorporate species suited to greater solar exposure, while deciduous species on northern elevations help preserve winter daylight.
- Glazing systems: Residential units use glazed openings behind the planted terraces. The vegetation and projecting balconies provide a secondary layer in front of the glazing, reducing direct solar exposure while maintaining views and access to daylight.
- Shading devices: Instead of conventional horizontal louvers or external blinds, trees, shrubs and projecting terraces form the primary shading system. The vegetation filters sunlight and creates variable shading depending on season, plant species and canopy density.
- Façade materiality: The solid façade uses dark grey porcelain stoneware cladding installed as a ventilated façade, supported on an aluminium substructure with stainless-steel brackets.
- Thermal Insulation: The planted façade provides an additional environmental buffer through shading, and increased insulation from external temperature variations.

II. ENVIRONMENTAL RESPONSE

- Solar Control: The vegetation acts as a dynamic solar-control layer. Tree canopies and shrubs filter solar radiation before it reaches the building façade and glazing.
- Daylight Optimization: Plant selection and distribution were carefully coordinated with façade orientation to prevent excessive obstruction of daylight.

- Natural Ventilation: The balconies and terraces create an intermediate outdoor zone between the apartments and the external environment. However, the primary environmental strategy is not dependent on façade-driven natural ventilation.
- Thermal comfort: The green second skin moderates the microclimate around the residential façade through shading and evapotranspiration.

III. PERFORMANCE

- Energy Efficiency: The green façade reduces energy demand through solar shading and improved thermal buffering.
- Solar radiation: Trees and shrubs intercept solar radiation before it reaches the building surface.
- Peak Thermal Loads: The dense summer vegetation significantly reduces solar heat gain and surface temperatures, helping lower cooling demand.

IV. INNOVATION: The key innovation is the transformation of the building façade into a functioning vertical ecosystem. Rather than treating vegetation as decoration added after the architectural design, Bosco Verticale integrates trees, shrubs and plants as an active component of the building envelope.

28.ParkRoyale Collection, Singapore



Project Overview:

- Location: 3 Upper Pickering Street, Singapore
- Architect: WOHA
- Completion: 2013
- Building Area: Approx. 29,811 m²
- Building Height: 12 storeys
- Major Functions: Hotel, offices, public areas, restaurants, swimming pools, sky gardens and landscaped terraces.
- Façade materials: Low-emissivity glazing, reinforced/precast concrete, projecting concrete sky-garden slabs and extensive landscape planting.

- Climate: Tropical rainforest climate
- Sustainability Certification: BCA Green Mark Platinum

I. ENVELOPE DESIGN:

- E-shaped tower configuration with projecting, curvilinear sky gardens integrated directly into the building envelope, creating a layered façade that combines enclosed hotel rooms with open landscaped terraces.
- Envelope Strategy: The building uses a combination of fully glazed guest-room façades and deep projecting sky gardens. The gardens act as an intermediate environmental layer between the interior spaces and the external tropical climate, while the podium incorporates undulating precast concrete layers and planted terraces.
- Orientation: The twelve-storey hotel tower follows an E-plan. Guest rooms are predominantly oriented towards the north, providing views towards Hong Lim Park.
- Glazing systems: The large glazed surfaces provide views towards the surrounding park and internal gardens while allowing substantial natural daylight into the rooms.
- Shading devices: The building is primarily self-shaded. The projecting sky gardens and deep horizontal concrete slabs shade the glazed façades.
- Façade materiality: The envelope combines dark reflective low-E glazing with projecting concrete sky-garden structures, extensive vegetation, water features and landscaped terraces.
- Thermal Insulation: The low-emissivity glazing reduces solar heat transfer through the extensive glazed areas, while the projecting garden slabs provide additional shading and reduce direct solar exposure.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar control is achieved primarily through the projecting sky gardens and their deep horizontal slabs. These create self-shading across the façades and reduce direct solar exposure to the low-E glazed rooms.
- Daylight Optimization: The E-shaped plan increases the amount of external façade available to the guest rooms, while the large glazed openings provide extensive daylight and views.
- Natural Ventilation: Naturally ventilated corridors are used as part of the passive environmental strategy. Open sky gardens and terraces provide outdoor spaces that remain connected to the tropical environment, while the building's open areas allow air movement through the circulation and landscaped zones.
- Thermal comfort: Thermal performance is supported through a combination of self-shading, low-E glazing, naturally ventilated circulation spaces and extensive vegetation.

III. PERFORMANCE

- Energy Efficiency: The building reduces cooling and lighting loads through passive measures including self-shading, low-E glazing, naturally ventilated corridors and extensive vegetation.
- Solar radiation: The projecting sky gardens form a major solar-control component of the envelope. Their horizontal projections intercept direct sunlight before it reaches the glazed rooms, reducing solar heat gain while maintaining views and daylight.
- Peak Thermal Loads: The combination of deep projecting terraces, vegetation and low-E glazing reduces direct solar exposure to the occupied spaces. The building's self-shading configuration is particularly suited to Singapore's high solar radiation and tropical climate.

IV. INNOVATION: The key innovation is the integration of landscape, façade and passive environmental design into a single building-envelope system. Rather than treating greenery as an independent landscape layer, WOHA incorporates large cantilevered sky gardens directly into the façade at multiple levels.

29.The Parcels Building, UK



Project Overview:

- Location: London, UK
- Architect: Grafton Architects, with TP Bennett
- Completion: 1957, redevelopment 2022
- Building Area: Approx. 5,450 m²
- Building Height: 6 storeys
- Major Functions: Mixed-use office and retail development, including flexible workspace, retail units, event space and rooftop terraces
- Façade materials: Portland Whitbread limestone, precast concrete, curtain walling, insulated glazing, aluminium framing and solar-coated glass.
- Climate: Temperate climate
- Sustainability Certification: BREEAM Outstanding

I. ENVELOPE DESIGN:

- The project replaces the outdated 1950s façade with a new fabric-first envelope while retaining 92.5% of the existing structure. Rather than demolishing the building, the new façade is attached directly to the retained structural frame through a bespoke bracket system.
- Envelope Strategy: The façade strategy combines adaptive reuse, high-performance curtain walling, solid stone fins, solar control, and improved insulation. This allowed the existing structure to be retained while substantially improving thermal and environmental performance
- Orientation: The façade design responds specifically to the building's two contrasting street orientations. The Oxford Street elevation receives additional depth and shading to reduce overheating
- Glazing systems: The new façade uses Schüco FWS 50 curtain walling, including FWS 50 SG systems and associated Schüco door systems. Continuous curtain walling provides improved airtightness and thermal performance compared with the original façade
- Shading devices: Deep stone fins and projecting façade elements provide external shading, particularly along Oxford Street. The depth of the stone façade was deliberately increased to respond to orientation and minimise internal overheating.
- Façade materiality: The envelope combines handset Portland Whitbread limestone, precast concrete and Schüco aluminium curtain walling with insulated, solar-coated glazing.
- Thermal Insulation: The new envelope substantially improves the thermal resistance of the original façade. The original façade was estimated at a U-value of approximately 5 W/m^2K, while the new wall system achieves approximately 0.3 W/m^2K.

II. ENVIRONMENTAL RESPONSE

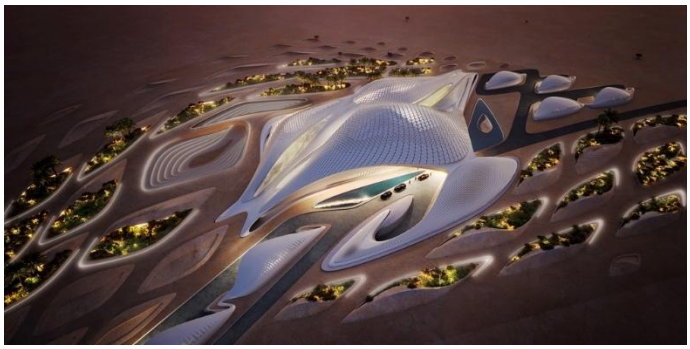
- Solar Control: The façade uses varying stone depths and projecting fins to provide external solar shading.
- Daylight Optimization: The new curtain-wall envelope allows significantly more natural light into the building than the original 1950s façade. The balance between increased glazing and solar control allows daylight penetration without creating excessive solar heat gain.
- Natural Ventilation: The primary environmental strategy is based on a high-performance, airtight envelope and energy-efficient building systems rather than relying predominantly on façade-driven natural ventilation.
- Thermal comfort: Improved insulation, airtightness, solar-coated glazing and external shading work together to reduce heat loss in winter and overheating in summer.

III. PERFORMANCE

- Energy Efficiency: The new façade significantly improves the thermal performance of the existing building.
- Solar radiation: The combination of deep stone fins, façade projections and solar-coated glazing reduces direct solar radiation entering the occupied spaces while retaining large areas of daylight-providing glazing.
- Peak Thermal Loads: External shading and solar-control glazing reduce unwanted solar gains and therefore help minimise summer cooling and overheating loads.

IV. INNOVATION: The key innovation is the development of a high-performance new façade around a retained 1950s structural frame, demonstrating how façade replacement can substantially improve building performance without demolishing the existing building.

30.BEEAH Headquarters, UAE



Project Overview:

- Location: Sharjah, UAE
- Architect: ZHA
- Completion: 2022
- Building Area: Approx. 9,000 m²
- Building Height: 18m
- Major Functions: BEEAH Group management and administrative headquarters, open-plan offices, smart meeting rooms, visitor centre and auditorium.
- Façade materials: Glass fibre reinforced concrete (GFRC) cladding panels, structural steel framing and extensive high-performance glazing.
- Climate: Hot desert climate
- Sustainability Certification: LEED Platinum

I. ENVELOPE DESIGN:

- Dune-inspired envelope composed of two primary interconnected building volumes that rise and curve like desert dunes, enclosing a central courtyard and creating a highly responsive building form.

- Envelope Strategy: The building envelope was developed as a series of interconnected “dunes” oriented and shaped according to local climatic conditions. The original concept of a continuous concrete shell was rationalized into a prefabricated steel structural frame clad with GFRC panels, allowing the complex geometry to be constructed efficiently.
- Orientation: The two primary dunes are positioned and shaped to respond to the harsh desert sun while providing daylight and views to internal spaces.
- Glazing systems: Glazing is selectively positioned rather than distributed uniformly across the façade. The design limits the amount of glass directly exposed to the harsh desert sun while providing daylight and views to occupied spaces.
- Shading devices: The primary shading strategy is integrated into the building geometry. The curved and inclined dune-like masses reduce direct solar exposure, while the limited glazing ratio minimizes unwanted solar heat gain.
- Façade materiality: Approximately 4,000 GFRC panels form the flowing white external skin. The panels are fixed to a prefabricated steel frame, creating the continuous curved surfaces that visually merge the roof and façade into the surrounding desert landscape.
- Thermal Insulation: The GFRC-clad envelope reduces solar heat gain, while slab cooling and glass cooling systems regulate internal temperatures.

II. ENVIRONMENTAL RESPONSE

- Solar Control: Solar control is achieved primarily through the orientation and geometry of the dune-shaped volumes, controlled glazing and GFRC external cladding.
- Daylight Optimization: The envelope is carefully designed to provide ample daylight to internal spaces without excessive exposure to direct solar radiation.
- Natural Ventilation: The central courtyard acts as an internal environmental zone and is integral to the natural ventilation strategy.
- Thermal comfort: Thermal comfort is achieved through the combination of passive envelope design and active cooling systems. The GFRC panels reduce solar gain, while slab and glass cooling regulate interior temperatures.

III. PERFORMANCE

- Energy Efficiency: The headquarters was designed to achieve net-zero energy consumption. Passive design strategies reduce the building's cooling demand, while an extensive photovoltaic system generates electricity on site.
- Solar radiation: The dune-like envelope minimizes direct exposure of glazed areas to the intense desert sun. GFRC cladding further reduces solar heat gain, while the photovoltaic system converts available solar radiation into renewable electricity.

- Peak Thermal Loads: Reducing exposed glazing, orienting the building volumes according to local solar conditions and using GFRC cladding together help reduce peak cooling loads.

IV. INNOVATION: The key innovation is the integration of climatically responsive parametric geometry, structural rationalization and high-performance façade design into a single envelope system. The original concrete-shell concept was transformed into a prefabricated steel frame carrying approximately 4,000 GFRC panels, allowing the complex dune-like geometry to be constructed efficiently while maintaining the architectural intent.

10.Criteria for Evaluating Case Study

Sr. No.	Parameter	0 Point awarded	1 Point awarded	2 Points awarded
1	Solar Control	No Clear Strategy	Basic shading/ Glazing control	Strong/ Integrated solar control Strategy
2	Thermal Performance	No Evidence	Some Insulation/ Glazing measures	Highly Integrated Thermal Envelope
3	Daylight	Poor/Not Considered	Some daylight consideration	Clearly optimized through envelope design
4	Natural Ventilation	Not Considered	Limited application	Strong passive ventilation strategy
5	Energy & Sustainability	No Clear Contribution	Some energy saving measures	Strong contribution to energy efficiency / green certification

11.Case study evaluation

Sr. No.	Case study	Solar Control	Thermal Performance	Daylight	Natural Ventilation	Energy & Sustainability	Total
1.	Case study 1	2	2	2	1	2	9/10
2.	Case study 2	2	1	2	0	1	6/10
3.	Case study 3	2	1	1	0	1	5/10
4.	Case study 4	2	2	2	0	1	7/10
5.	Case study 5	2	1	2	1	1	7/10
6.	Case study 6	2	2	2	0	1	7/10
7.	Case study 7	2	0	1	2	1	6/10
8.	Case study 8	2	1	1	0	1	5/10
9.	Case study 9	1	2	2	0	2	7/10
10.	Case study 10	2	2	2	1	2	9/10
11.	Case study 11	2	2	2	2	2	10/10
12.	Case study 12	2	1	2	0	2	7/10
13.	Case study 13	2	2	2	0	2	8/10
14.	Case study 14	1	1	2	0	1	5/10
15.	Case study 15	2	2	2	0	1	7/10

16.	Case study 16	2	2	2	0	2	8/10
17.	Case study 17	2	2	2	0	2	8/10
18.	Case study 18	1	1	1	0	1	4/10
19.	Case study 19	2	2	2	1	2	9/10
20.	Case study 20	2	2	2	0	2	8/10
21.	Case study 21	2	2	2	2	2	10/10
22.	Case study 22	2	2	2	1	2	9/10
23.	Case study 23	2	2	2	1	2	9/10
24.	Case study 24	2	2	2	2	2	10/10
25.	Case study 25	2	1	2	0	2	7/10
26.	Case study 26	1	2	2	1	1	7/10
27.	Case study 27	2	2	2	1	2	9/10
28.	Case study 28	2	2	2	2	2	10/10
29.	Case study 29	2	2	2	0	2	8/10
30.	Case study 30	2	2	2	2	2	10/10

12.Initial Analysis

Most of the case studies demonstrate strong envelope performance, but the performance is uneven across the five parameters.

The strongest recurring strategies are solar control and daylight, while natural ventilation is the weakest parameter across the overall sample. This shows that Envelope Integration is more important than individual components.

Key Performance Indicators

Sr. no.	Indicator	Result
1	Total case studies	30
2	Max. possible score	300
3	Average score	7.7/10
4	Average performance	77%
5	Case studies scoring 8/10	16/30
6	Case studies scoring 9/10	11/30
7	Case studies scoring 10/10	5/30

Parameter Performance

Sr. No.	Parameter	Total score / 60	Average /2	Performance
1	Solar control	56	1.87	93%
2	Thermal performance	50	1.67	83%
3	Daylight	56	1.87	93%
4	Natural Ventilation	20	0.67	33%
5	Energy & sustainability	49	1.63	82%

Top performing case studies:

1. Sunstar Headquarters
2. Leeza Soho
3. The Black & White Building
4. Parkroyale Collection
5. BEEAH Headquarters

These projects achieve the maximum score because their envelope strategies address multiple environmental parameters simultaneously, rather than solving only solar control or thermal performance

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